THE UNIVERSITY OF MELBOURNE

Student number

Semester 1 Assessment, 2025

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 25 pages (including this page) with 8 questions and 90 total marks

Permitted Materials

- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. There is extra space you can use for answers to any question commencing on page 22. If you still do not have enough space, tick the box near the bottom of page 25 and request extra pages from an invigilator—include the question number at the top of each additional page.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.

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Question 1 (9 marks)

Let p and q be primitive statements.

(a) Write a truth table for the compound statement

$$q \implies (p \vee (\sim q)).$$

(b) If q is true and $q \implies (p \vee (\sim q))$ is true, what can you conclude about the truth value of p? Justify your answer in terms of specific lines in truth table from Part (a).

(c) Find a statement with **exactly one connective** $\sim, \wedge, \vee, \implies$, or $\iff$ which is logically equivalent to $q \implies (p \vee (\sim q))$. Justify your answer using a truth table.

(a)

p	q	$\sim q$	$p \vee (\sim q)$	$q \implies (p \vee (\sim q))$
T	T	F	T	T
T	F	T	T	T
F	T	F	F	F
F	F	T	T	T

Correct truth table.

(b) By Line 1 of the truth table from Part (a), if q and $q \implies (p \vee (\sim q)$ are both true, then p must be True.

Correct answer that p must be True.

Correct justification that Line 1 refers to the situation where q and $q \implies (p \vee (\sim q)$ are both true.

(c) $q \implies (p \vee (\sim q)$ is logically equivalent to $q \implies p$. Using Part (a)

p	q	$q \implies (p \vee (\sim q))$	$q \implies p$
T	T	T	T
T	F	T	T
F	T	F	F
F	F	T	T

p	q	$[q \implies (p \vee (\sim q))] \iff [q \implies p]$
T	T	T
T	F	T
F	T	T
F	F	T

Since $[q \implies ((p \vee (\sim q))] \iff [q \implies p]$ is a tautology, $q \implies ((p \vee (\sim q))$ is logically equivalent to $q \implies p$.

Correct answer that $q \implies ((p \vee (\sim q))$ is logically equivalent to $q \implies p$.

Correct truth table (which is allowed to use Part (a)).

Justification for why $[q \implies ((p \vee (\sim q))] \equiv [q \implies p]$.

Ideally, the solution consists of showing $[q \implies ((p \vee (\sim q))] \iff [q \implies p]$ is a tautology as above. However, noting that $q \implies ((p \vee (\sim q))$ and $q \implies p$ take the same truth values is okay for full marks.

Question 2 (9 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \setminus \{0\} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Using only the Axioms of the Real Numbers (and rules of logic) together with the fact that $0x = 0$ for all $x \in \mathbb{R}$, write a (semi-)formal proof showing that for all $a, b \in \mathbb{R}$

$$\text{if } a > 0 \text{ and } b < 0, \text{ then } ab < 0.$$

Proof.

1. Let $a \in \mathbb{R}$ such that $a > 0$.	(premise)
2. Let $b \in \mathbb{R}$ such that $b < 0$.	(premise)
3. $\exists(-b) \in \mathbb{R} \;\; b + (-b) = 0$	(A5)
4. $b + (-b) < 0 + (-b)$	(2., OA)
5. $0 < 0 + (-b)$	(3., 4.)
6. $0 < (-b) + 0$	(5., A2)
7. $0 < -b$	(6., A4)
8. $0 < a(-b)$	(1., 7., OM)
9. $0 + ab < a(-b) + ab$	(8., OA)
10. $ab + 0 < ab + a(-b)$	(9., A2)
11. $ab < ab + a(-b)$	(10., A4)
12. $ab < a\,(b + (-b))$	(11., D)
13. $ab < a \cdot 0$	(2., 12.)
14. $ab < 0 \cdot a$	(13., M2)
15. $ab < 0$	(14., $0a = 0$)

Proof is of the form of a formal proof.
Overall correct proof (separate from the justification).
Correct justification for each step using the Axioms of the Real Numbers.
Including using the Axioms carefully.

Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 3 (9 marks)

Let A and B be subsets of $\mathbb{R}$ which are both non-empty and bounded above. Define

$$A + B = \{x + y : x \in A \text{ and } y \in B\}.$$

(a) Show that if α is an upper bound of A and β is an upper bound of B, then $\alpha + \beta$ is an upper bound of $A + B$.

(b) Show that $\sup(A + B) = \sup A + \sup B$.

(a) Let $z \in A + B$. Then $z = x + y$ for some $x \in A$ and $y \in B$. Since α is an upper bound of A, $x \leq \alpha$. Since β is an upper bound of B, $y \leq \beta$. Hence

$$z = x + y \leq \alpha + \beta.$$

Therefore, for all $z \in A + B$, $z \leq \alpha + \beta$. Therefore, $\alpha + \beta$ is an upper bound of $A + B$.

Argues that for an arbitrary $z \in A + B$, $z \leq \alpha + \beta$.

Any correct proof that $\alpha + \beta$ is an upper bound of $A + B$.

(b) By Part (a), $\sup A + \sup B$ is an upper bound of $A + B$.

Let $\epsilon > 0$. Since $\sup A - \epsilon/2$ is not an upper bound of A, there exists $x \in A$ such that $x > \sup A - \epsilon/2$. Since $\sup B - \epsilon/2$ is not an upper bound of B, there exists $y \in B$ such that $y > \sup B - \epsilon/2$. Hence $z = x + y \in A + B$ and

$$z = x + y > \sup A + \sup B - \epsilon.$$

Therefore, for every $\epsilon > 0$ there exists $z \in A + B$ such that $z > \sup A + \sup B - \epsilon$. Therefore, $\sup A + \sup B$ is the least upper bound of $A + B$.

States that by Part (a), $\sup A + \sup B$ is an upper bound of $A + B$.

Argues that $\sup A + \sup B$ is the least upper bound of $A + B$, e.g. for an arbitrary $\epsilon > 0$ there exists $z \in A + B$ such that $z \leq \alpha + \beta - \epsilon$.

Any correct proof that $\sup A + \sup B$ is the least upper bound of $A + B$.

Question 4 (12 marks)

For each of the following situations, provide a specific example for which the situation is true, and briefly justify your answer.

(a) A set $E \subseteq \mathbb{Q}$ such that the set of all limit points of E is $[0, \infty)$.

(b) A sequence (f_n) such that (f_n^2) converges to 9 but (f_n) diverges.
(Here $f_n^2 = f_n \cdot f_n$ for each $n \in \mathbb{N}^+$.)

(c) A convergent series $\sum_{n=1}^{\infty} a_n$ such that $\sum_{n=1}^{\infty} a_n^2$ diverges.

(a) Let $E = [0, \infty) \cap \mathbb{Q}$.

Let $a \in [0, \infty)$. Let $\delta > 0$. There exists a rational number $x \in (a, a+\delta) \subset (a-\delta, a+\delta) \setminus \{a\}$. Since $x > a \geq 0$ and x is a rational number, $x \in E$. Therefore, there exists $x \in E \cap (a-\delta, a+\delta) \setminus \{a\}$. Therefore, a is a limit point of E.

Let $a \in (-\infty, 0)$. Let $\delta = 1 - a > 0$. Then $E \cap (a-\delta, a+\delta) = \emptyset$. Therefore, a is not a limit point of E. Typo: this delta doesn't work (can you see why?). However, delta = -a/2 > 0 works for this argument.

Any correct answer for the set E.

Correct justification. Must explain why each $a \in [0, \infty)$ is a limit point of E and why each $a \in (-\infty, 0)$ is not a limit point of E.

(b) Let $f_n = 3(-1)^n$ for each $n \in \mathbb{N}^+$.

$f_n^2 = 9$ for all $n \in \mathbb{N}^+$. Since (f_n^2) is a constant sequence, $f_n^2 \to 9$.

$f_{2k-1} = -3$ and $f_{2k} = 3$ for all $k \in \mathbb{N}^+$. Since (f_{2k-1}) and (f_{2k}) are both constant sequences, $f_{2k-1} \to -3$ and $f_{2k} \to 3$. Hence (f_n) has two subsequences with different limits. By the Divergence Criterion, (f_n) diverges.

Any correct answer for the sequence (f_n).

Correct justification. Must explain why (f_n^2) converges to 9 but (f_n) diverges.

(c) Let $a_n = \dfrac{(-1)^n}{\sqrt{n}}$ for all $n \in \mathbb{N}^+$.

Since $\frac{1}{\sqrt{n}}$ is decreasing in n and $\frac{1}{\sqrt{n}} \to 0$ as $n \to \infty$,

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$$

converges.

But by the p-series test

$$\sum_{n=1}^{\infty} a_n^2 = \sum_{n=1}^{\infty} \frac{1}{n}$$

diverges.

Any correct answer for the series $\displaystyle\sum_{n=1}^{\infty} a_n$.

Correct justification. Must explain why $\displaystyle\sum_{n=1}^{\infty} a_n$ converges and why $\displaystyle\sum_{n=1}^{\infty} a_n^2$ diverges.

Question 5 (13 marks)

Let (x_n) be the sequence such that $x_1 = 0$ and

$$x_{n+1} = \frac{9}{6 - x_n}$$

for each $n \in \mathbb{N}^+$.

(a) Using a proof by induction, show that $0 \leq x_n < 3$ for all $n \in \mathbb{N}^+$.

(b) Is the sequence (x_n) monotone increasing, monotone decreasing, or neither? Justify your answer.

(Hint: Compute $x_{n+1} - x_n$.)

(c) Is the sequence (x_n) Cauchy? Briefly justify your answer.

(a) We want to show that

$$p(n) \ : \ 0 \leq x_n < 3$$

is true for all $n \in \mathbb{N}^+$. We shall use induction on n.

Base case. Show that $p(1)$ is true.
When $n = 1$, $x_1 = 0$ and thus $0 \leq x_1 < 3$.

Inductive step. Show that for each $k \in \mathbb{N}^+$, $p(k) \implies p(k+1)$.
Suppose that for some $k \in \mathbb{N}^+$

$$0 \leq x_k < 3.$$

Then

$$3 < 6 - x_k \leq 6$$
$$\frac{3}{2} \leq x_{k+1} = \frac{9}{6 - x_k} < 3.$$

Therefore, $0 \leq x_k < 3$.

Conclusion. Therefore, by the Principle of Mathematical Induction, $0 \leq x_n < 3$ for all $n \in \mathbb{N}^+$.

Proof of the base case $n = 1$.

Proof of the inductive step.

States the conclusion "Therefore, by the Principle of Mathematical Induction, $0 \leq x_n < 3$ for all $n \in \mathbb{N}^+$."

(b) Yes, (x_n) is monotone increasing.

For each integer $n \in \mathbb{N}^+$

$$\begin{aligned} x_{n+1} - x_n &= \frac{9}{6 - x_n} - x_n \\ &= \frac{9 - 6x_n + x_n^2}{6 - x_n} \\ &= \frac{(x_n - 3)^2}{6 - x_n} > 0. \end{aligned}$$

Therefore, $x_n < x_{n+1}$ for all $n \in \mathbb{N}^+$. That is, (x_n) is monotone increasing.

Correctly states that (x_n) is monotone increasing.

Attempts to verify that $x_n < x_{n+1}$.

Correct computation that $x_n < x_{n+1}$.

(c) Yes, (x_n) is Cauchy.

By Part (a), (x_n) is bounded. By Part (b), (x_n) is monotone increasing. Hence (x_n) converges. Since (x_n) converges, (x_n) is Cauchy.

Correct answer that (x_n) is Cauchy.

Any complete and correct explanation for why (x_n) is Cauchy.

Question 6 (11 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 2x - 4 & \text{if } x < 2 \\ \dfrac{x}{3x+1} & \text{if } x \geq 2 \end{cases}$$

(a) Using the $\epsilon - M$ definition of limits, show that $\lim_{x\to\infty} f(x) = 1/3$.

(b) Using the $\epsilon - \delta$ definition of continuity, show that f is not continuous at 2.

(a) Let $\epsilon > 0$. Let $M = \max\left\{2, \frac{1}{9\epsilon} - \frac{1}{3}\right\}$. Let $x \in \mathbb{R}$ such that $x > M$. Then

$$\begin{aligned} \frac{1}{9\epsilon} - \frac{1}{3} &< x \\ \frac{1}{3\epsilon} &< 3x + 1 \\ \frac{1}{3(3x+1)} &< \epsilon \\ \left|\frac{-1}{3(3x+1)}\right| &< \epsilon \\ \left|\frac{3x - (3x+1)}{3(3x+1)}\right| &< \epsilon \\ \left|\frac{x}{3x+1} - \frac{1}{3}\right| &< \epsilon \end{aligned}$$

Since $x > M \geq 2$

$$\left|f(x) - \frac{1}{3}\right| < \epsilon.$$

Therefore, if $x \in \mathbb{R}$ such that $x > M$ then $|f(x) - 1/3| < \epsilon$. Therefore, $\lim_{x\to\infty} f(x) = 1/3$.

Proof considers an arbitrary $\epsilon > 0$, chooses $M \in \mathbb{R}$, and considers an arbitrary $x > M$.

Any correct value for M.

Any correct argument that if $x > M$ then $|f(x) - 1/3| < \epsilon$.

Asserts the conclusion that "Therefore, if $x > M$ then $|f(x) - 1/3| < \epsilon$. Therefore, $\lim_{x\to\infty} f(x) = 1/3$.

(b) Let $\epsilon = 2/7$. Let $\delta > 0$. Let $x = 2 - \delta/2$. Then

$$|x - 2| = |(2 - \delta/2) - 2| = |-\delta/2| = \delta/2 < \delta$$

and

$$|f(x) - f(2)| = \left|(2x - 4) - \frac{2}{7}\right| = \left|-\delta - \frac{2}{7}\right| = \delta + \frac{2}{7} \geq \frac{2}{7}.$$

Therefore, for every $\delta > 0$ there exists $x \in \mathbb{R}$ such that $|x - 2| < \delta$ and $|f(x) - f(2)| \geq 2/7$. Therefore, f is not continuous at 2.

Any correct choice of ϵ.

Any correct choice of x.

Correct argument that $|x - 2| < \delta$.

Correct argument that $|f(x) - f(2)| \geq \epsilon$.

Asserts the conclusion that "Therefore, for every $\delta > 0$ there exists $x \in \mathbb{R}$ such that $|x - 2| < \delta$ and $|f(x) - f(2)| \geq \epsilon$. Therefore, f is not continuous at 2.

Question 7 (13 marks)

Let $f : [-1, 1] \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 1 - x^2 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

(a) Let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be any partition of $[-1, 1]$. Compute the lower Riemann sum $L(f, P)$.

(b) Find a positive lower bound for the upper Riemann integral of f on $[-1, 1]$.

(c) Is f Riemann integrable on $[-1, 1]$? Justify your answer using Parts (a) and (b).

(a) Since each subinterval $[x_{i-1}, x_i]$ contains irrational numbers

$$m_i = \inf_{x \in [x_{i-1}, x_i]} f(x) = \inf_{x \in [x_{i-1}, x_i]} 0 = 0.$$

Hence

$$\begin{aligned} L(f, P) &= \sum_{i=1}^{n} m_i (x_i - x_{i-1}) \\ &= \sum_{i=1}^{n} 0 \cdot (x_i - x_{i-1}) \\ &= 0 \end{aligned}$$

In Parts (a) and (b), mentions that each sub-interval contains irrational and rational numbers.

Correctly computes $m_i = 0$.

Correctly computes $L(f, P) = 0$.

(b) Let $P = \{x_0, x_1, x_2, \ldots, x_n\}$ be any partition of $[-1, 1]$. By replacing P with a refinement we may assume that $-1/2, 1/2 \in P$. Since $f(x) \geq 0$ on each subinterval $[x_{i-1}, x_i]$

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x) \geq 0.$$

Let $[x_{i-1}, x_i]$ be an subinterval contained in $[-1/2, 1/2]$. Notice that $[x_{i-1}, x_i]$ contains infinitely many rational numbers x, at which $f(x) = 1 - x^2 \geq 1 - (1/2)^2 = 3/4$. Thus

$$M_i = \sup_{x \in [x_{i-1}, x_i]} f(x) \geq 3/4.$$

Now letting $x_{i_1} = -1/2$ and $x_{i_2} = 1/2$

$$\begin{aligned} U(f, P) &= \sum_{i=1}^{n} M_i (x_i - x_{i-1}) \\ &\geq \sum_{i=i_1+1}^{i_2} \frac{3}{4}(x_i - x_{i-1}) \\ &= \frac{3}{4}(x_{i_1+1} - x_{i_1}) + \frac{3}{4}(x_{i_1+2} - x_{i_1+1}) + \cdots + \frac{3}{4}(x_{i_2} - x_{i_2-1}) \\ &= \frac{3}{4}(x_{i_2} - x_{i_1}) = \frac{3}{4} \cdot 1 = \frac{3}{4}. \end{aligned}$$

Since $3/4$ is a lower bound for the set of all upper Riemann sums $U(f, P)$, $U(f) \geq 3/4$.

Any correct positive lower bound for $U(f)$.

Correctly computes a lower bound on M_i for each subinterval.

Correctly computes the lower bound on $U(f, P)$.

Finds a lower bound for $U(f)$ using the definition $U(f) = \inf\{U(f, P) \mid P \in \widetilde{\mathcal{P}}\}$.

(c) No, f is not Riemann integrable.

By Part (b),

$$L(f) = \sup\{L(f, P) \mid P \in \widetilde{\mathcal{P}}\} = \sup\{0\} = 0.$$

By Part (c), $U(f) \geq 3/4$. Since

$$L(f) = 0 < 3/4 \leq U(f),$$

f is not Riemann integrable.

Correct answer that f is not Riemann integrable.

Correct justification using Parts (a) and (b).

Note. *For Question 8, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 8 (14 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=1}^{\infty} \frac{2n}{n^4+16}$

(ii) $\displaystyle\sum_{n=2}^{\infty} \frac{(-1)^n}{n \log n}$

(b) Find the radius of convergence and interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{3^n x^n}{\sqrt{n(n+1)}}$$

(a) (i) Converges.

Method 1. By the p-series test,

$$\sum_{n=1}^{\infty} \frac{1}{n^3}$$

converges. Since $0 \leq \frac{2n}{n^4+16} \leq \frac{2}{n^3}$ for all $n \in \mathbb{N}^+$, by the Direct Comparison Test,

$$\sum_{n=1}^{\infty} \frac{2n}{n^4+16}$$

converges.

Method 2. We compute that

$$\int_1^{\infty} \frac{2x}{x^4+16}\,dx = \int_1^{\infty} \frac{du}{u^2+16} = \left[\tfrac{1}{4}\arctan(u/4)\right]_1^{\infty} = \frac{\pi}{8} < \infty.$$

Correct answer that the sum converges.

Correct justification using a test for convergence of an infinite series.

(ii) Converges.

Since $\log(n)$ is an increasing function of n, $\frac{1}{n\log n}$ is an decreasing function of n. Moreover

$$\lim_{n\to\infty} \frac{1}{n\log n} = 0.$$

Hence by the Alternating Series Test, the series converges.

Correct answer that the sum converges.

Correct justification using a test for convergence of an infinite series (such as the Alternating Series Test).

(b) We have that

$$\begin{aligned}\lim_{n\to\infty} \frac{|a_{n+1}|}{|a_n|} &= \lim_{n\to\infty} \frac{3^{n+1}|x|^{n+1}}{\sqrt{(n+1)(n+2)}} \cdot \frac{\sqrt{n(n+1)}}{3^n x^n} \\ &= \lim_{n\to\infty} \frac{3\sqrt{n}\,|x|}{\sqrt{n+1}} \\ &= 3|x|.\end{aligned}$$

By the Limit Ratio Test, the series converges whenever $3|x| < 1$, i.e. $|x| < 1/3$. Similarly, the series diverges whenever $|x| > 1/3$. Therefore, the radius of convergence is $1/3$.

At the endpoint $x = -1/3$, the power series is

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n(n+1)}}$$

Since the sequence $\left(\frac{1}{\sqrt{n(n+1)}}\right)$ is monotone decreasing and converges to zero, the power series converges at $x = -1/3$.

At the endpoint $x = 1/3$, the series is

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n(n+1)}}$$

By direct comparison with the p-series $\sum_{n=1}^{\infty} \frac{1}{n}$, which diverges, the series diverges at $x = 1/3$.

Therefore, the interval of convergence is $[-1/3, 1/3)$.

Correctly answers that the radius of convergence is $1/3$.

Correctly uses the Limit Ratio Test to find the radius of convergence.

Correct argument for why the series converges at $x = -1/3$.

Correct argument for why the series diverges at $x = 1/3$.

Correctly answers that the interval of convergence is $[-1/3, 1/3)$.

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End of Exam — Total Available Marks = 90

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